HOW RELIABLE IS LEARNED CARDINALITY ESTIMATION?

A Data Model-Centric Analysis and Empirical Study

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WHY RELIABILITY?

Cardinality estimates steer plan choice in cost-based optimizers; errors can lead to much slower plans. Improved average accuracy does not by itself ensure reliability: low median errors can coexist with severe error tails under data and workload shifts. We assess reliability along three dimensions:

Directional bias
Systematic over- or underestimation.
Measured by the underestimation rate

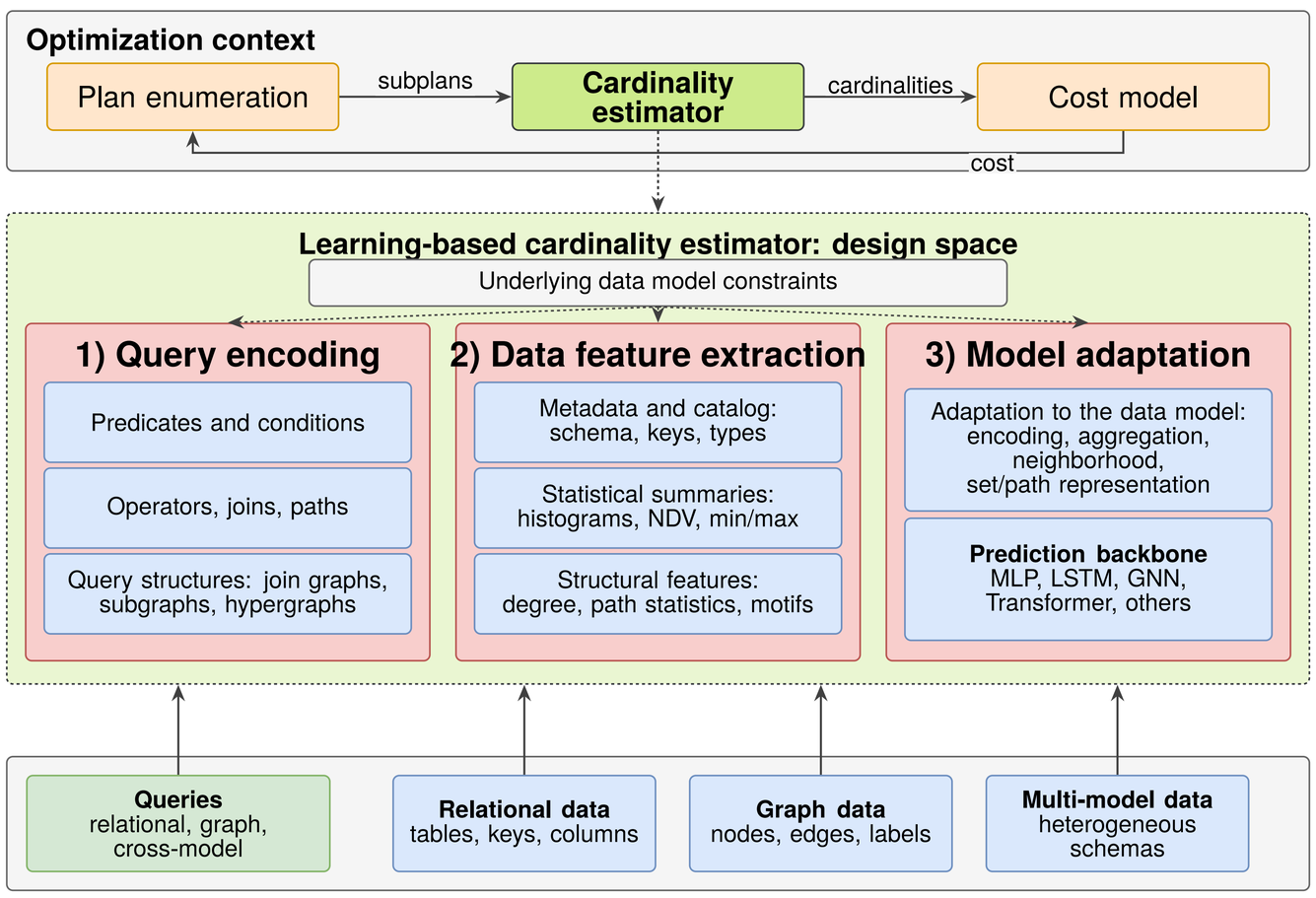
Upper-tail risk
Severe errors on a minority of queries.
Measured by upper q-error percentiles (P95, P99)

Shift degradation
Loss of accuracy when data or workloads change.
Measured by change in these statistics across workloads

Error metric: q-error $= \max(\hat{c}/c,\ c/\hat{c})$, with estimate $\hat{c}$ and truth c.

A DATA MODEL-CENTRIC FRAMEWORK

Three analytical dimensions describe a learned estimator; the underlying data model constrains each of them.



THE AXIS SHIFTS

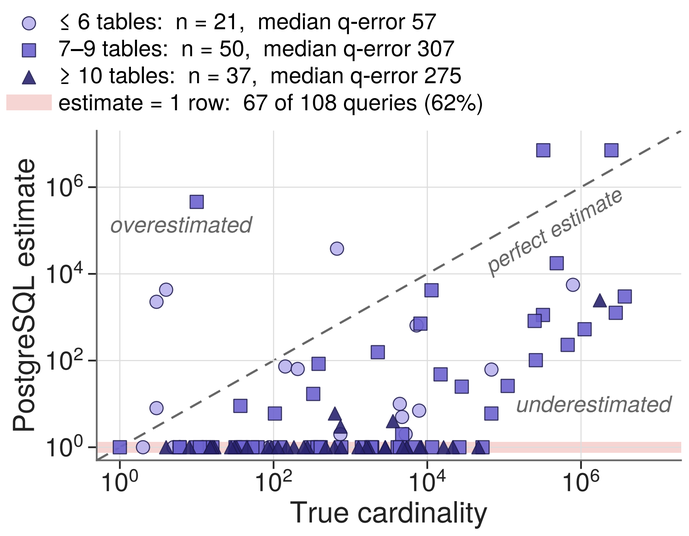
The most informative way to classify estimators differs by data model.

	Primary modeling target
Relational	*Distribution learning:* NeuroCard, DeepDB, FACE, FactorJoin *Query regression:* MSCN, JGMP, ALECE, GRASP
	Topology-handling strategy
Graph	*Direct structural encoding:* GNCE, NeurSC, FlowSC *Path-sequence encoding:* SPACE *Substructure decomposition:* ALSS
	Cross-model integration strategy
Multi-model	*Pre-estimation bridging:* LVE, Qi et al. *Joint learned estimation:* MANN *Post-estimation composition:* open route

Why it shifts. Data models differ in *data-side modelability* and *query-side structural complexity* (qualitative). Harder dependencies and richer query structure force estimators to choose what to encode, simplify, or decompose.

POSTGRESQL BIAS

PostgreSQL 18.1, JOB (108 of 113 queries nonempty), estimate of the topmost join node

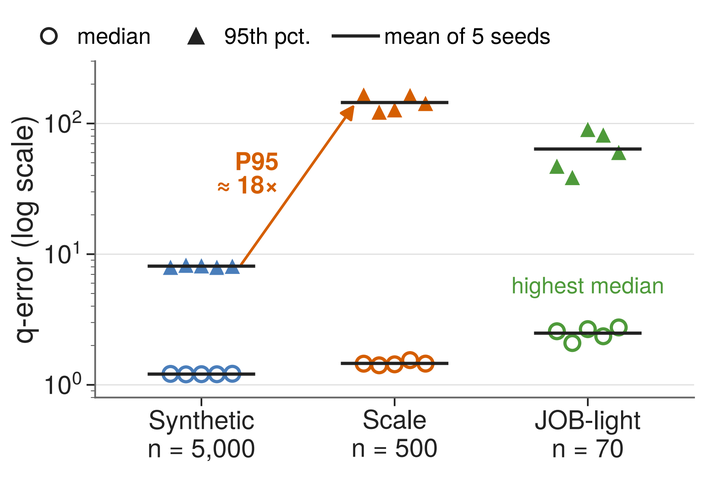


92.6% underestimated
261 median q-error
>14k 95th pct.

Median q-error is approximately five times higher for queries with more than six joined tables, with no clear further increase for wider joins. The ≥10-table group also spans the GEQO threshold.

MSCN WORKLOAD SHIFT

Five seeds (41–45); one trained checkpoint per seed, evaluated on all three workloads without retraining

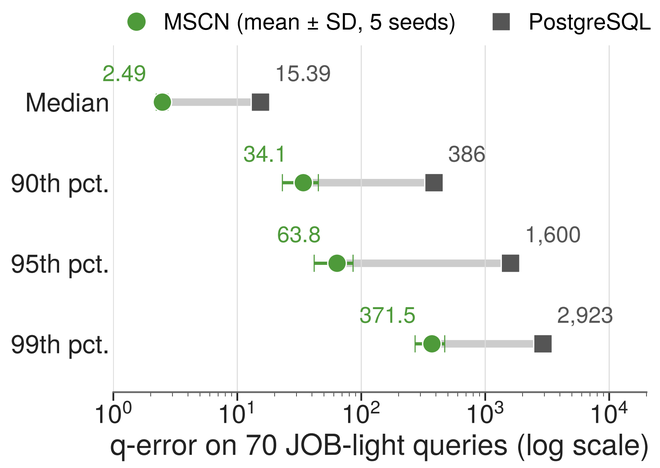


≈18× P95, Scale vs. Synthetic
2.49 JOB-light median (highest)

Shifts change the *kind* of failure: JOB-light raises typical error, while Scale (larger join counts) inflates the tail. The ordering holds for every seed.

MATCHED JOB-LIGHT

Same 70 JOB-light queries for both estimators



PostgreSQL underestimates 69 of 70 queries. MSCN cuts typical and tail error substantially, but its remaining tail shows that failures are reduced, not eliminated.

Not resource-matched. MSCN used 90,000 labeled training queries; PostgreSQL used default statistics. No claim about deployment cost or plan quality.

TAKEAWAYS

- Reliability is a joint property of the **estimator**, the **workload**, and the **data model**.
- Report bias, tail percentiles, seed variation, and controlled shifts, not only median q-error.
- Graph and multi-model CE lack benchmarks that isolate estimator reliability.

Paper & Resources

OPEN CHALLENGES

- Reliability-oriented evaluation protocols
- Adaptation without fresh labels
- Pretrained and zero-shot estimation
- LLM-generated workloads, semantic operators

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